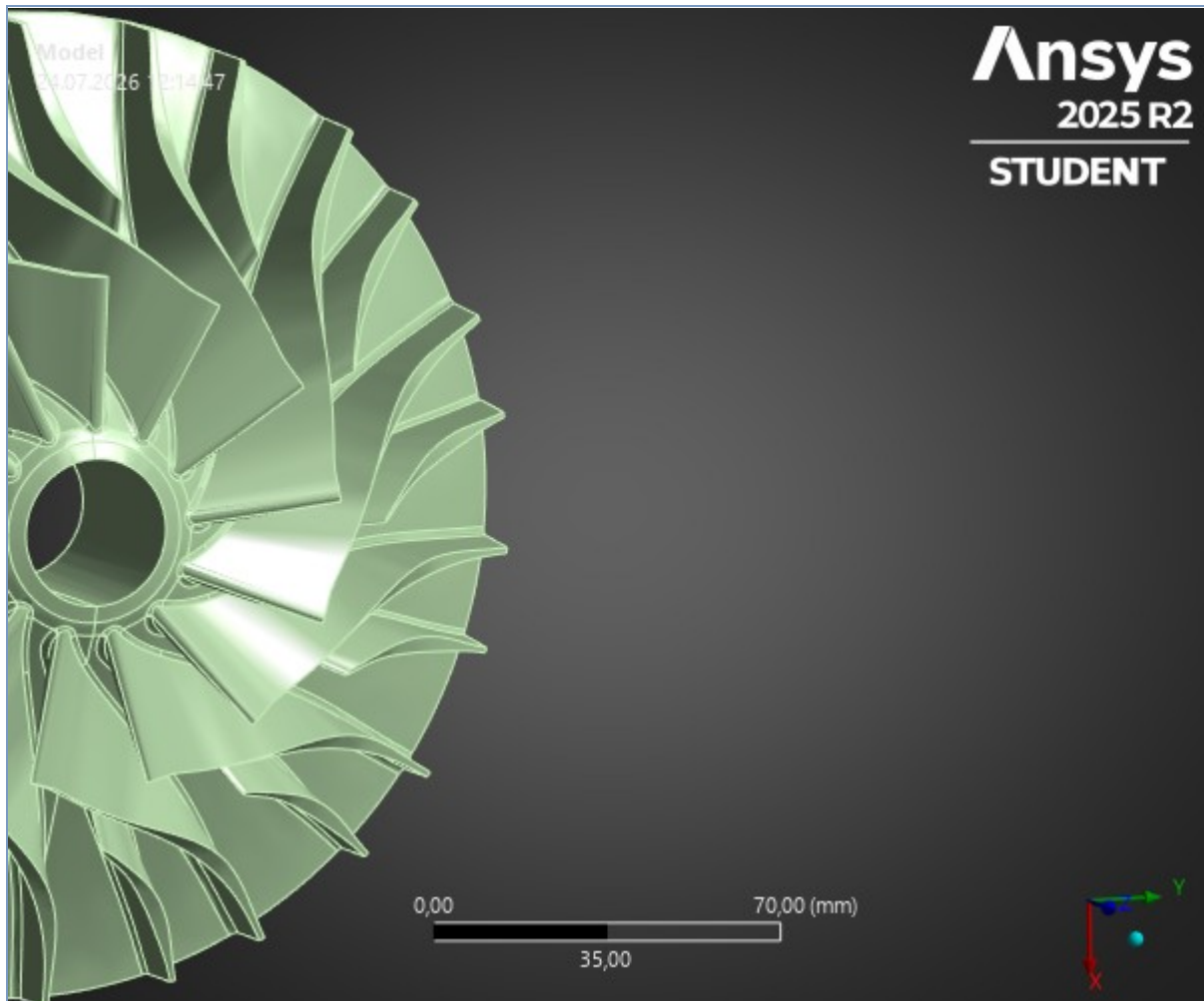




## Project\*

|                              |                         |
|------------------------------|-------------------------|
| First Saved                  | Monday, July 20, 2026   |
| Last Saved                   | Thursday, July 23, 2026 |
| Product Version              | 2025 R2                 |
| Save Project Before Solution | No                      |
| Save Project After Solution  | No                      |



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## Units

**TABLE 1**

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| Unit System         | Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius |
| Angle               | Degrees                                             |
| Rotational Velocity | rad/s                                               |
| Temperature         | Celsius                                             |

## Model (B4)

**TABLE 2**

**Model (B4) > Geometry Imports**

|             |                         |
|-------------|-------------------------|
| Object Name | <i>Geometry Imports</i> |
| State       | Solved                  |

**TABLE 3**

**Model (B4) > Geometry Imports > Geometry Import (B3)**

|                               |                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------|
| Object Name                   | <i>Geometry Import (B3)</i>                                                     |
| State                         | Solved                                                                          |
| <b>Definition</b>             |                                                                                 |
| Source                        | C:\Users\User\Desktop\Прочностные характеристики\max_pressure\max_pressure.step |
| Type                          | Step                                                                            |
| <b>Basic Geometry Options</b> |                                                                                 |
| Solid Bodies                  | Yes                                                                             |
| Surface Bodies                | Yes                                                                             |

|                                   |             |
|-----------------------------------|-------------|
| Line Bodies                       | No          |
| Parameters                        | Independent |
| Parameter Key                     | ANS;DS      |
| Attributes                        | No          |
| Named Selections                  | No          |
| Material Properties               | No          |
| <b>Advanced Geometry Options</b>  |             |
| Use Associativity                 | Yes         |
| Coordinate Systems                | No          |
| Reader Mode Saves Updated File    | No          |
| Use Instances                     | Yes         |
| Smart CAD Update                  | Yes         |
| Compare Parts On Update           | No          |
| Analysis Type                     | 3-D         |
| Mixed Import Resolution           | None        |
| Import Facet Quality              | Source      |
| Clean Bodies On Import            | No          |
| Stitch Surfaces On Import         | None        |
| Decompose Disjoint Geometry       | Yes         |
| Enclosure and Symmetry Processing | Yes         |

## Geometry

**TABLE 4**  
**Model (B4) > Geometry**

|                               |                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------|
| Object Name                   | <i>Geometry</i>                                                                 |
| State                         | Fully Defined                                                                   |
| <b>Definition</b>             |                                                                                 |
| Source                        | C:\Users\User\Desktop\Прочностные характеристики\max_pressure\max_pressure.step |
| Type                          | Step                                                                            |
| Length Unit                   | Millimeters                                                                     |
| Element Control               | Program Controlled                                                              |
| Display Style                 | Body Color                                                                      |
| <b>Bounding Box</b>           |                                                                                 |
| Length X                      | 200, mm                                                                         |
| Length Y                      | 200, mm                                                                         |
| Length Z                      | 51,025 mm                                                                       |
| <b>Properties</b>             |                                                                                 |
| Volume                        | 6,9549e+005 mm <sup>3</sup>                                                     |
| Mass                          | 0,72331 kg                                                                      |
| Scale Factor Value            | 1,                                                                              |
| <b>Statistics</b>             |                                                                                 |
| Bodies                        | 1                                                                               |
| Active Bodies                 | 1                                                                               |
| Nodes                         | 127598                                                                          |
| Elements                      | 72770                                                                           |
| Mesh Metric                   | None                                                                            |
| <b>Update Options</b>         |                                                                                 |
| Assign Default Material       | No                                                                              |
| <b>Basic Geometry Options</b> |                                                                                 |
| Solid Bodies                  | Yes                                                                             |
| Surface Bodies                | Yes                                                                             |
| Line Bodies                   | No                                                                              |

| Parameters                              | Independent |
|-----------------------------------------|-------------|
| Parameter Key                           | ANS;DS      |
| Attributes                              | No          |
| Named Selections                        | No          |
| Material Properties                     | No          |
| <b>Advanced Geometry Options</b>        |             |
| Use Associativity                       | Yes         |
| Coordinate Systems                      | No          |
| Reader Mode Saves Updated File          | No          |
| Use Instances                           | Yes         |
| Smart CAD Update                        | Yes         |
| Compare Parts On Update                 | No          |
| Analysis Type                           | 3-D         |
| Mixed Import Resolution                 | None        |
| Import Facet Quality                    | Source      |
| Clean Bodies On Import                  | No          |
| Stitch Surfaces On Import               | None        |
| Decompose Disjoint Geometry             | Yes         |
| ID_GeometryPrefProcessPhysicsDefinition | No          |
| Enclosure and Symmetry Processing       | Yes         |

**TABLE 5**  
**Model (B4) > Geometry > Parts**

|                            |                              |
|----------------------------|------------------------------|
| Object Name                | max_pressure Вырез-Вытянуть2 |
| State                      | Meshed                       |
| <b>Graphics Properties</b> |                              |
| Visible                    | Yes                          |
| Transparency               | 1                            |
| <b>Definition</b>          |                              |
| Suppressed                 | No                           |
| Stiffness Behavior         | Flexible                     |
| Coordinate System          | Default Coordinate System    |
| Reference Temperature      | By Environment               |
| Treatment                  | None                         |
| <b>Material</b>            |                              |
| Assignment                 | PA12-CF10                    |
| Nonlinear Effects          | Yes                          |
| Thermal Strain Effects     | Yes                          |
| <b>Bounding Box</b>        |                              |
| Length X                   | 200, mm                      |
| Length Y                   | 200, mm                      |
| Length Z                   | 51,025 mm                    |
| <b>Properties</b>          |                              |
| Volume                     | 6,9549e+005 mm <sup>3</sup>  |
| Mass                       | 0,72331 kg                   |
| Centroid X                 | -4,5014e-003 mm              |
| Centroid Y                 | 9,8156e-004 mm               |
| Centroid Z                 | 13,043 mm                    |
| Moment of Inertia Ip1      | 1361,4 kg·mm <sup>2</sup>    |
| Moment of Inertia Ip2      | 1361, kg·mm <sup>2</sup>     |
| Moment of Inertia Ip3      | 2513,2 kg·mm <sup>2</sup>    |
| <b>Statistics</b>          |                              |
| Nodes                      | 127598                       |
| Elements                   | 72770                        |
| Mesh Metric                | None                         |

**TABLE 6**  
**Model (B4) > Materials**

|                      |                  |
|----------------------|------------------|
| Object Name          | <i>Materials</i> |
| State                | Fully Defined    |
| <b>Statistics</b>    |                  |
| Materials            | 4                |
| Material Assignments | 0                |

## Coordinate Systems

**TABLE 7**  
**Model (B4) > Coordinate Systems > Coordinate System**

|                            |                                 |
|----------------------------|---------------------------------|
| Object Name                | <i>Global Coordinate System</i> |
| State                      | Fully Defined                   |
| <b>Definition</b>          |                                 |
| Type                       | Cartesian                       |
| Coordinate System ID       | 0,                              |
| <b>Origin</b>              |                                 |
| Origin X                   | 0, mm                           |
| Origin Y                   | 0, mm                           |
| Origin Z                   | 0, mm                           |
| <b>Directional Vectors</b> |                                 |
| X Axis Data                | [ 1, 0, 0, ]                    |
| Y Axis Data                | [ 0, 1, 0, ]                    |
| Z Axis Data                | [ 0, 0, 1, ]                    |
| <b>Transfer Properties</b> |                                 |
| Source                     |                                 |
| Read Only                  | No                              |

## Connections

**TABLE 8**  
**Model (B4) > Connections**

|                                          |                    |
|------------------------------------------|--------------------|
| Object Name                              | <i>Connections</i> |
| State                                    | Fully Defined      |
| <b>Auto Detection</b>                    |                    |
| Generate Automatic Connection On Refresh | Yes                |
| <b>Transparency</b>                      |                    |
| Enabled                                  | Yes                |
| <b>Statistics</b>                        |                    |
| Contacts                                 | 0                  |
| Active Contacts                          | 0                  |
| Joints                                   | 0                  |
| Active Joints                            | 0                  |
| Beams                                    | 0                  |
| Active Beams                             | 0                  |
| Bearings                                 | 0                  |
| Active Bearings                          | 0                  |
| Springs                                  | 0                  |
| Active Springs                           | 0                  |
| Body Interactions                        | 0                  |
| Active Body Interactions                 | 0                  |

## Mesh

**TABLE 9**

| <b>Model (B4) &gt; Mesh</b>              |                       |
|------------------------------------------|-----------------------|
| Object Name                              | <i>Mesh</i>           |
| State                                    | Solved                |
| <b>Display</b>                           |                       |
| Display Style                            | Use Geometry Setting  |
| <b>Defaults</b>                          |                       |
| Physics Preference                       | Mechanical            |
| Element Order                            | Program Controlled    |
| Element Size                             | 3,39 mm               |
| <b>Sizing</b>                            |                       |
| Use Adaptive Sizing                      | Yes                   |
| Resolution                               | Default (2)           |
| Mesh Defeaturing                         | Yes                   |
| Defeature Size                           | Default               |
| Transition                               | Fast                  |
| Span Angle Center                        | Coarse                |
| Initial Size Seed                        | Assembly              |
| Bounding Box Diagonal                    | 287,41 mm             |
| Average Surface Area                     | 531,5 mm <sup>2</sup> |
| Minimum Edge Length                      | 7,3636e-004 mm        |
| <b>Quality</b>                           |                       |
| Check Mesh Quality                       | Yes, Errors           |
| Error Limits                             | Aggressive Mechanical |
| Target Element Quality                   | Default (5,e-002)     |
| Smoothing                                | Medium                |
| Mesh Metric                              | None                  |
| <b>Inflation</b>                         |                       |
| Use Automatic Inflation                  | None                  |
| Inflation Option                         | Smooth Transition     |
| Transition Ratio                         | 0,272                 |
| Maximum Layers                           | 5                     |
| Growth Rate                              | 1,2                   |
| Inflation Algorithm                      | Pre                   |
| Inflation Element Type                   | Wedges                |
| View Advanced Options                    | No                    |
| <b>Advanced</b>                          |                       |
| Number of CPUs for Parallel Part Meshing | Program Controlled    |
| Straight Sided Elements                  | No                    |
| Rigid Body Behavior                      | Dimensionally Reduced |
| Triangle Surface Mesher                  | Program Controlled    |
| Topology Checking                        | Yes                   |
| Pinch Tolerance                          | Please Define         |
| Generate Pinch on Refresh                | No                    |
| Auto-Map Fillets                         | No                    |
| <b>Automatic Methods</b>                 |                       |
| Sheet Body Method                        | Prime Quad Dominant   |
| Sweepable Body Method                    | Sweep                 |
| <b>Statistics</b>                        |                       |
| Nodes                                    | 127598                |
| Elements                                 | 72770                 |
| Show Detailed Statistics                 | No                    |

**TABLE 10**  
**Model (B4) > Mesh > Mesh Controls**

| Object Name  | <i>Face Sizing</i> |
|--------------|--------------------|
| State        | Fully Defined      |
| <b>Scope</b> |                    |

|                   |                    |
|-------------------|--------------------|
| Scoping Method    | Geometry Selection |
| Geometry          | 78 Faces           |
| <b>Definition</b> |                    |
| Suppressed        | No                 |
| Type              | Element Size       |
| Element Size      | 2,8 mm             |
| <b>Advanced</b>   |                    |
| Defeature Size    | Default            |
| Influence Volume  | No                 |
| Behavior          | Soft               |

## Named Selections

## Static Structural (B5)

**TABLE 11**  
**Model (B4) > Analysis**

|                         |                               |
|-------------------------|-------------------------------|
| Object Name             | <i>Static Structural (B5)</i> |
| State                   | Solved                        |
| <b>Definition</b>       |                               |
| Physics Type            | Structural                    |
| Analysis Type           | Static Structural             |
| Solver Target           | Mechanical APDL               |
| <b>Options</b>          |                               |
| Environment Temperature | 22, °C                        |
| Generate Input Only     | No                            |

**TABLE 12**  
**Model (B4) > Static Structural (B5) > Analysis Settings**

|                               |                          |
|-------------------------------|--------------------------|
| Object Name                   | <i>Analysis Settings</i> |
| State                         | Fully Defined            |
| <b>Step Controls</b>          |                          |
| Number Of Steps               | 1,                       |
| Current Step Number           | 1,                       |
| Step End Time                 | 1, s                     |
| Auto Time Stepping            | On                       |
| Define By                     | Substeps                 |
| Initial Substeps              | 20,                      |
| Minimum Substeps              | 10,                      |
| Maximum Substeps              | 100,                     |
| <b>Solver Controls</b>        |                          |
| Solver Type                   | Program Controlled       |
| Weak Springs                  | Program Controlled       |
| Solver Pivot Checking         | Program Controlled       |
| Large Deflection              | On                       |
| Inertia Relief                | Off                      |
| Quasi-Static Solution         | On                       |
| <b>Rotordynamics Controls</b> |                          |
| Coriolis Effect               | Off                      |
| <b>Restart Controls</b>       |                          |
| Generate Restart Points       | Program Controlled       |
| Retain Files After Full Solve | No                       |
| Combine Restart Files         | Program Controlled       |
| <b>Nonlinear Controls</b>     |                          |
| Newton-Raphson Option         | Program Controlled       |

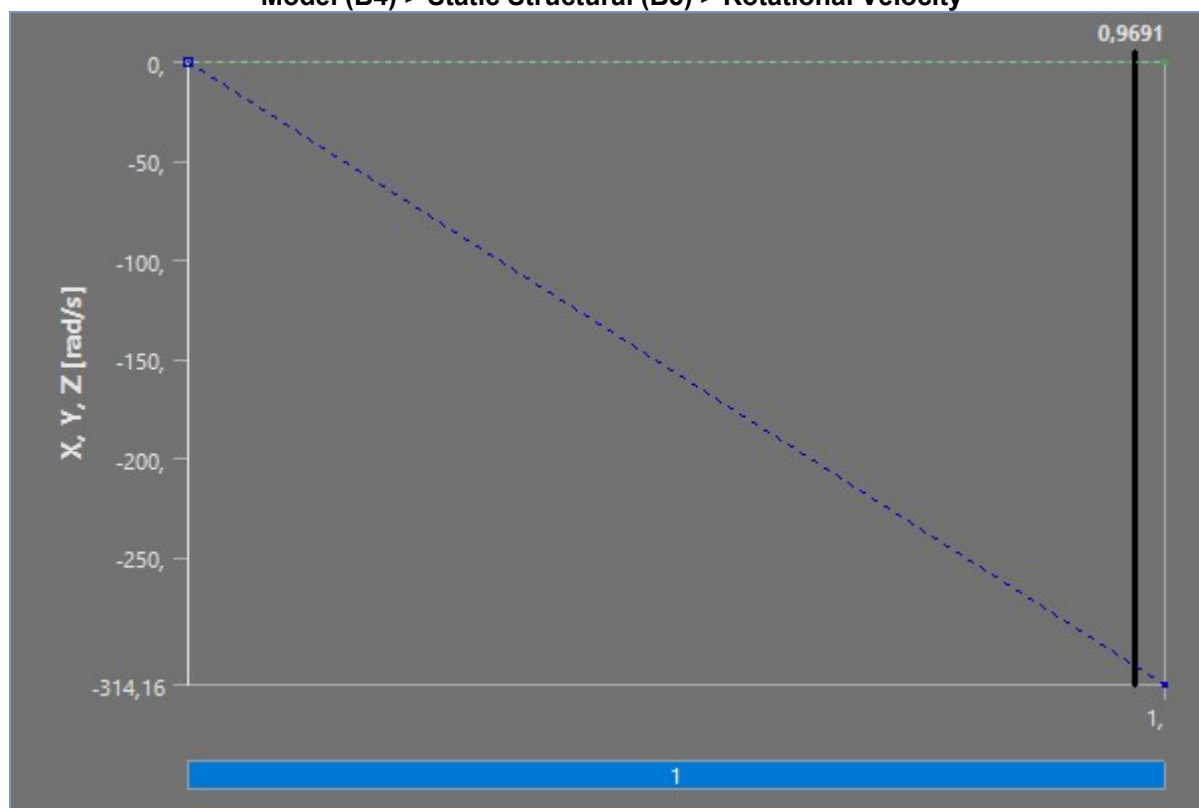
|                                 |                                                                                          |
|---------------------------------|------------------------------------------------------------------------------------------|
| Force Convergence               | Program Controlled                                                                       |
| Moment Convergence              | Program Controlled                                                                       |
| Displacement Convergence        | Program Controlled                                                                       |
| Rotation Convergence            | Program Controlled                                                                       |
| Line Search                     | Program Controlled                                                                       |
| Stabilization                   | Constant                                                                                 |
| --Method                        | Energy                                                                                   |
| --Energy Dissipation Ratio      | 1,e-004                                                                                  |
| --Activation For First Substep  | No                                                                                       |
| --Stabilization Force Limit     | 0,2                                                                                      |
| <b>Advanced</b>                 |                                                                                          |
| Inverse Option                  | No                                                                                       |
| Contact Split (DMP)             | Program Controlled                                                                       |
| <b>Output Controls</b>          |                                                                                          |
| Output Selection                | None                                                                                     |
| Stress                          | Yes                                                                                      |
| Back Stress                     | No                                                                                       |
| Strain                          | Yes                                                                                      |
| Contact Data                    | Yes                                                                                      |
| Nonlinear Data                  | No                                                                                       |
| Nodal Forces                    | No                                                                                       |
| Volume and Energy               | Yes                                                                                      |
| Euler Angles                    | Yes                                                                                      |
| General Miscellaneous           | No                                                                                       |
| Contact Miscellaneous           | No                                                                                       |
| Store Results At                | All Time Points                                                                          |
| Result File Compression         | Program Controlled                                                                       |
| <b>Analysis Data Management</b> |                                                                                          |
| Solver Files Directory          | C:\Users\User\Desktop\Прочностные характеристики\All_work_pressure_files\dp0\SYS-1\MECH\ |
| Future Analysis                 | None                                                                                     |
| Scratch Solver Files Directory  |                                                                                          |
| Save MAPDL db                   | No                                                                                       |
| Contact Summary                 | Program Controlled                                                                       |
| Delete Unneeded Files           | Yes                                                                                      |
| Nonlinear Solution              | Yes                                                                                      |
| Solver Units                    | Active System                                                                            |
| Solver Unit System              | nmm                                                                                      |

**TABLE 13**  
**Model (B4) > Static Structural (B5) > Rotations**

|                   |                            |
|-------------------|----------------------------|
| Object Name       | <i>Rotational Velocity</i> |
| State             | Fully Defined              |
| <b>Scope</b>      |                            |
| Scoping Method    | Geometry Selection         |
| Geometry          | All Bodies                 |
| <b>Definition</b> |                            |
| Define By         | Components                 |
| Coordinate System | Global Coordinate System   |
| X Component       | 0, rad/s (ramped)          |
| Y Component       | 0, rad/s (ramped)          |
| Z Component       | -314,16 rad/s (ramped)     |

|              |                 |
|--------------|-----------------|
| X Coordinate | 5,e-004 mm      |
| Y Coordinate | -6,1961e-004 mm |
| Z Coordinate | 4,65e-002 mm    |
| Suppressed   | No              |

**FIGURE 1**  
**Model (B4) > Static Structural (B5) > Rotational Velocity**

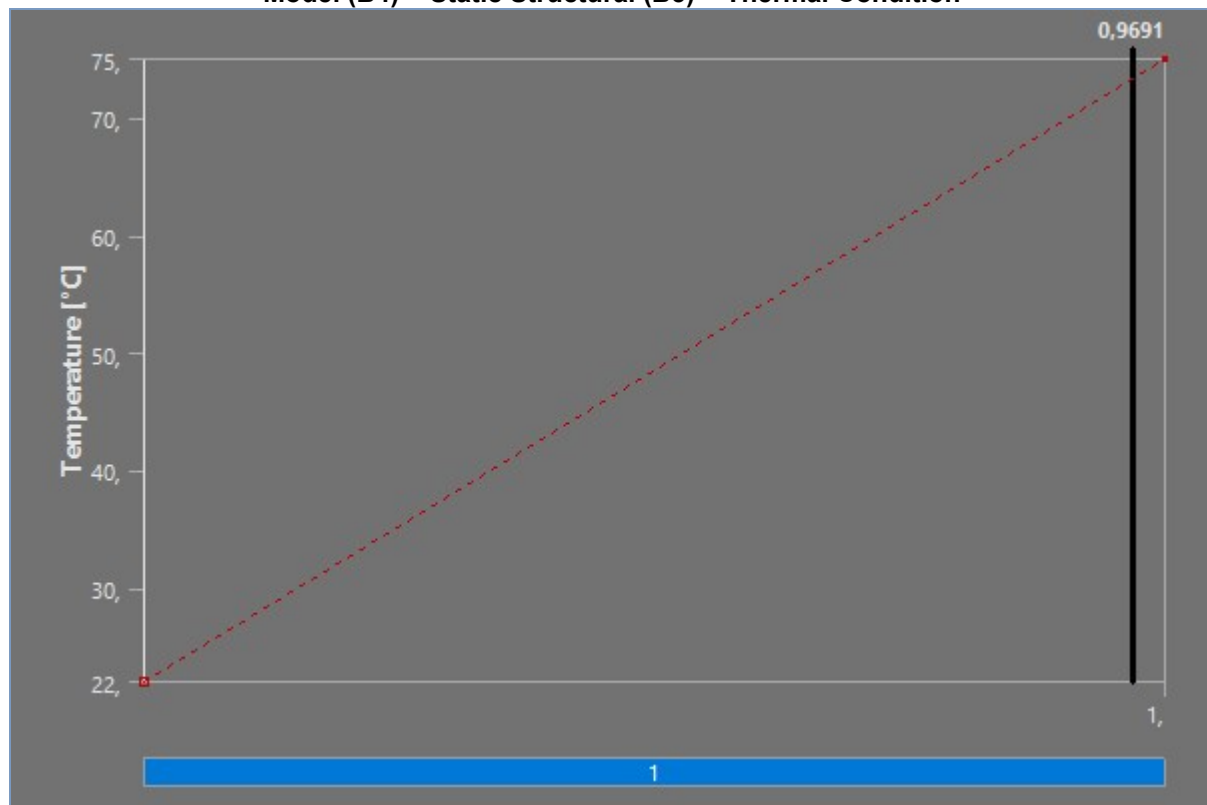


**TABLE 14**  
**Model (B4) > Static Structural (B5) > Loads**

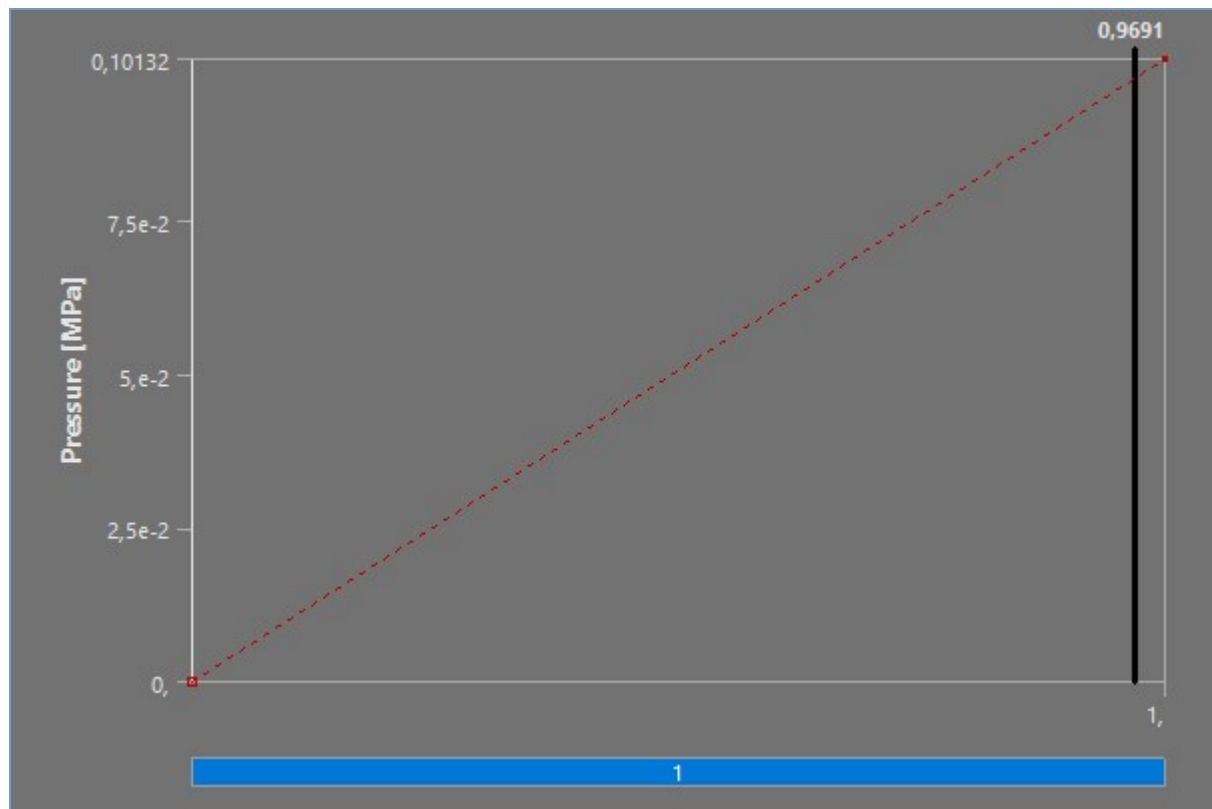
| Object Name       | Thermal Condition  | Pressure             | Cylindrical Support | Remote Displacement      |
|-------------------|--------------------|----------------------|---------------------|--------------------------|
| State             | Fully Defined      |                      |                     | Suppressed               |
| Scope             |                    |                      |                     |                          |
| Scoping Method    | Geometry Selection |                      |                     |                          |
| Geometry          | 1 Body             | 78 Faces             | 2 Faces             |                          |
| Coordinate System |                    |                      |                     | Global Coordinate System |
| X Coordinate      |                    |                      |                     | 2,7818e-016 mm           |
| Y Coordinate      |                    |                      |                     | 2,6427e-016 mm           |
| Z Coordinate      |                    |                      |                     | 21,75 mm                 |
| Location          |                    |                      |                     | Defined                  |
| Definition        |                    |                      |                     |                          |
| Type              | Thermal Condition  | Pressure             | Cylindrical Support | Remote Displacement      |
| Magnitude         | 75, °C (ramped)    | 0,10132 MPa (ramped) |                     |                          |
| Suppressed        | No                 |                      |                     | Yes                      |
| Define By         |                    | Normal To            |                     |                          |
| Applied By        |                    | Surface Effect       |                     |                          |
| Loaded Area       |                    | Deformed             |                     |                          |
| Radial            |                    |                      | Fixed               |                          |
| Axial             |                    |                      | Fixed               |                          |
| Tangential        |                    |                      | Free                |                          |

|                 |  |                |
|-----------------|--|----------------|
| X Component     |  | 0, mm (ramped) |
| Y Component     |  | 0, mm (ramped) |
| Z Component     |  | 0, mm (ramped) |
| Rotation X      |  | 0, ° (ramped)  |
| Rotation Y      |  | 0, ° (ramped)  |
| Rotation Z      |  | 0, ° (ramped)  |
| Behavior        |  | Deformable     |
| <b>Advanced</b> |  |                |
| Pinball Region  |  | All            |

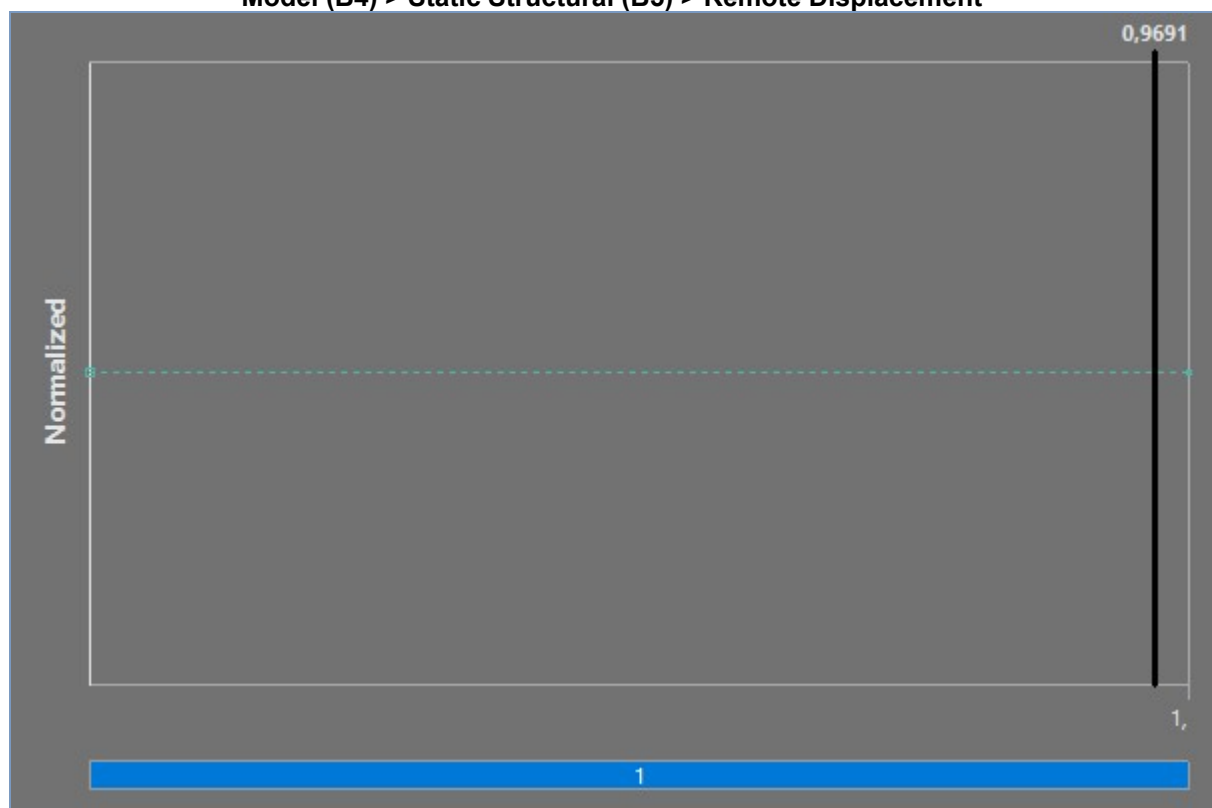
**FIGURE 2**  
**Model (B4) > Static Structural (B5) > Thermal Condition**



**FIGURE 3**  
**Model (B4) > Static Structural (B5) > Pressure**



**FIGURE 4**  
**Model (B4) > Static Structural (B5) > Remote Displacement**



### ***Solution (B6)***

**TABLE 15**  
**Model (B4) > Static Structural (B5) > Solution**

| Object Name | <i>Solution (B6)</i> |
|-------------|----------------------|
| State       | Solved               |

| Adaptive Mesh Refinement |           |
|--------------------------|-----------|
| Max Refinement Loops     | 1,        |
| Refinement Depth         | 2,        |
| Information              |           |
| Status                   | Done      |
| MAPDL Elapsed Time       | 6 m 6 s   |
| MAPDL Memory Used        | 935, MB   |
| MAPDL Result File Size   | 525,19 MB |
| Post Processing          |           |
| Beam Section Results     | No        |
| On Demand Stress/Strain  | No        |

**TABLE 16**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Solution Information**

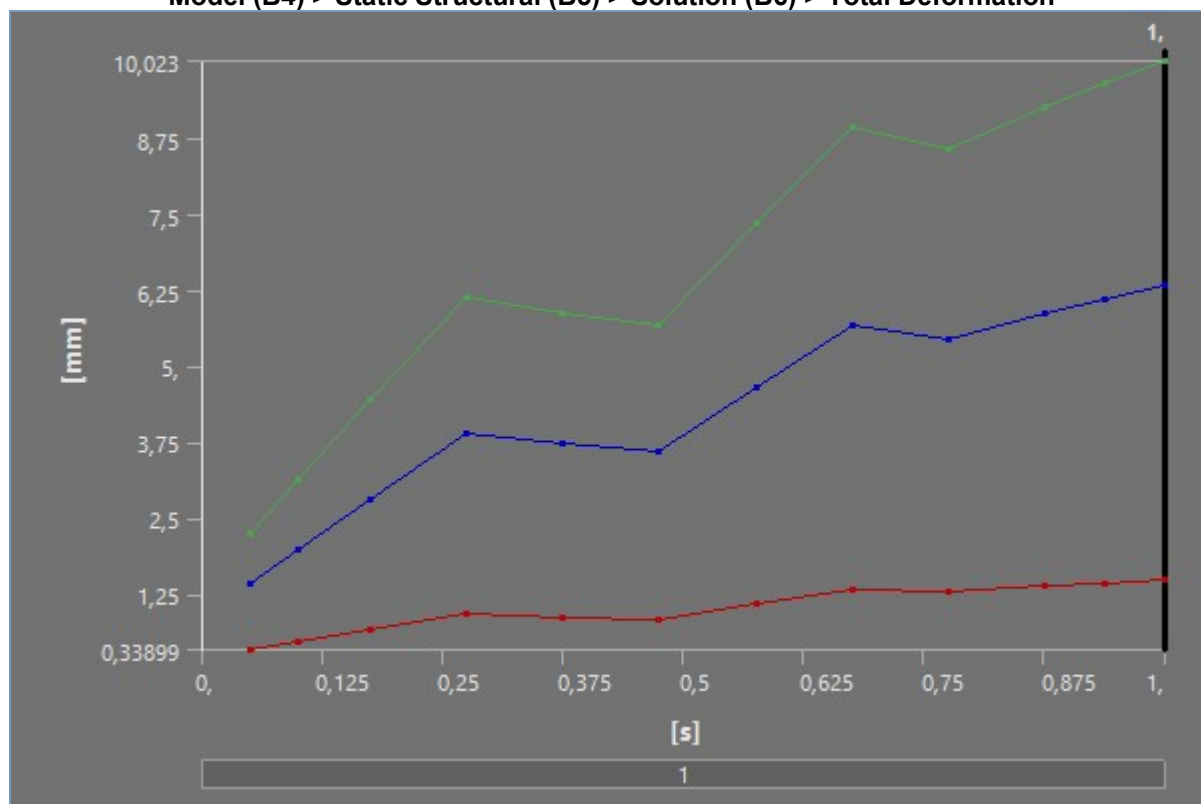
| Object Name                  | <i>Solution Information</i> |
|------------------------------|-----------------------------|
| State                        | Solved                      |
| Solution Information         |                             |
| Solution Output              | Solver Output               |
| Newton-Raphson Residuals     | 0                           |
| Identify Element Violations  | 5                           |
| Update Interval              | 2,5 s                       |
| Display Points               | All                         |
| FE Connection Visibility     |                             |
| Activate Visibility          | Yes                         |
| Display                      | All FE Connectors           |
| Draw Connections Attached To | All Nodes                   |
| Line Color                   | Connection Type             |
| Visible on Results           | No                          |
| Line Thickness               | Single                      |
| Display Type                 | Lines                       |

**TABLE 17**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Results**

|                                                               |                              |                               |
|---------------------------------------------------------------|------------------------------|-------------------------------|
| Model (24) - Static Structural (25) - Solution (26) - Results |                              |                               |
| Object Name                                                   | Total Deformation            | Equivalent Stress             |
| State                                                         | Solved                       |                               |
| Scope                                                         |                              |                               |
| Scoping Method                                                | Geometry Selection           |                               |
| Geometry                                                      | All Bodies                   |                               |
| Definition                                                    |                              |                               |
| Type                                                          | Total Deformation            | Equivalent (von-Mises) Stress |
| By                                                            | Time                         |                               |
| Display Time                                                  | Last                         |                               |
| Separate Data by Entity                                       | No                           |                               |
| Calculate Time History                                        | Yes                          |                               |
| Identifier                                                    |                              |                               |
| Suppressed                                                    | No                           |                               |
| Results                                                       |                              |                               |
| Minimum                                                       | 1,4938 mm                    | 1,042e-003 MPa                |
| Maximum                                                       | 10,023 mm                    | 20,897 MPa                    |
| Average                                                       | 6,3488 mm                    | 0,57006 MPa                   |
| Minimum Occurs On                                             | max_pressure Вырез-Вытянуть2 |                               |
| Maximum Occurs On                                             | max_pressure Вырез-Вытянуть2 |                               |
| Minimum Value Over Time                                       |                              |                               |
| Minimum                                                       | 0,33899 mm                   | 8,7779e-005 MPa               |
| Maximum                                                       | 1,4938 mm                    | 3,7179e-003 MPa               |
| Maximum Value Over Time                                       |                              |                               |
|                                                               |                              |                               |

|                           |           |            |
|---------------------------|-----------|------------|
| Minimum                   | 2,2612 mm | 2,2704 MPa |
| Maximum                   | 10,023 mm | 20,897 MPa |
| Information               |           |            |
| Time                      | 1, s      |            |
| Load Step                 | 1         |            |
| Substep                   | 12        |            |
| Iteration Number          | 39        |            |
| Integration Point Results |           |            |
| Display Option            |           | Averaged   |
| Average Across Bodies     |           | No         |

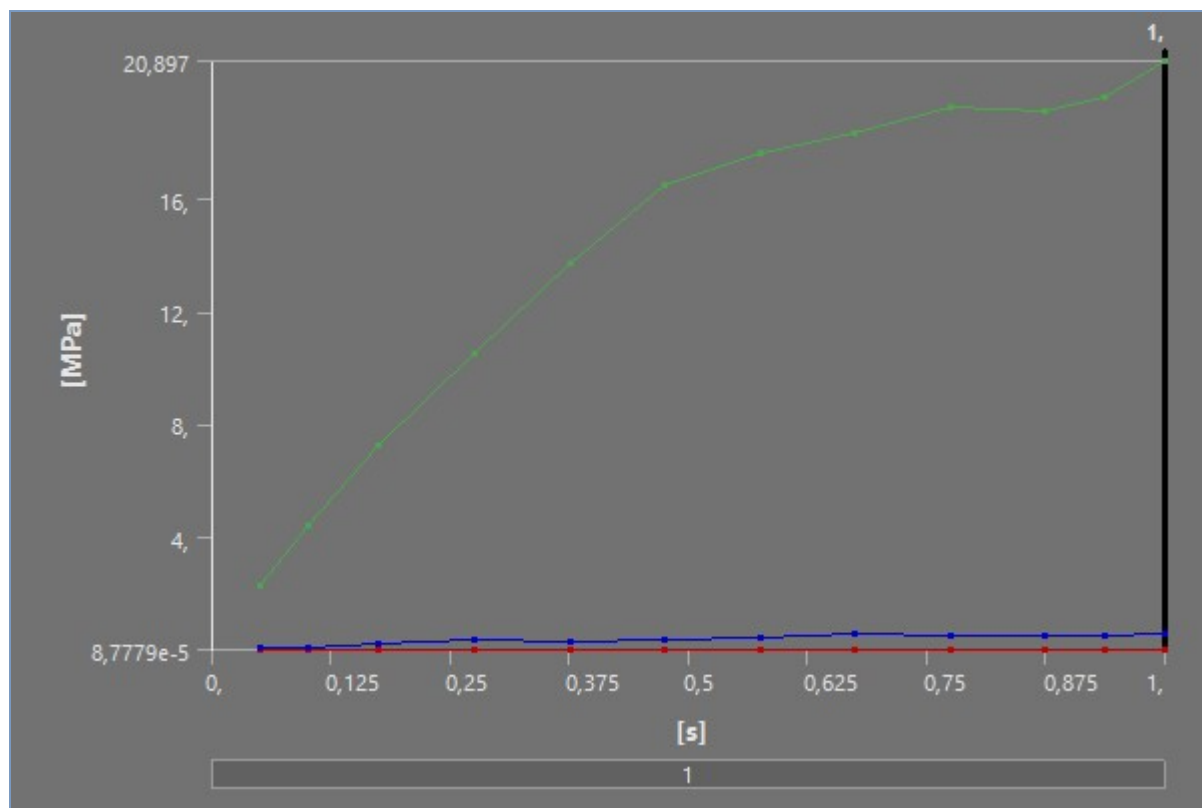
**FIGURE 5**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation**



**TABLE 18**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|----------|--------------|--------------|--------------|
| 5,e-002  | 0,33899      | 2,2612       | 1,4337       |
| 0,1      | 0,47096      | 3,1426       | 1,9925       |
| 0,175    | 0,66649      | 4,4499       | 2,821        |
| 0,275    | 0,91934      | 6,1435       | 3,8934       |
| 0,375    | 0,87856      | 5,8726       | 3,7227       |
| 0,475    | 0,84704      | 5,6652       | 3,592        |
| 0,575    | 1,0972       | 7,3431       | 4,654        |
| 0,675    | 1,3336       | 8,9348       | 5,6597       |
| 0,775    | 1,2802       | 8,578        | 5,4355       |
| 0,875    | 1,3808       | 9,258        | 5,8654       |
| 0,9375   | 1,4402       | 9,6598       | 6,1192       |
| 1,       | 1,4938       | 10,023       | 6,3488       |

**FIGURE 6**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress**

**TABLE 19****Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|----------|---------------|---------------|---------------|
| 5,e-002  | 8,7779e-005   | 2,2704        | 5,4435e-002   |
| 0,1      | 1,2769e-004   | 4,4122        | 0,1031        |
| 0,175    | 7,2797e-004   | 7,2708        | 0,19157       |
| 0,275    | 3,2839e-003   | 10,521        | 0,34187       |
| 0,375    | 2,7726e-004   | 13,7          | 0,31084       |
| 0,475    | 2,1565e-003   | 16,491        | 0,33679       |
| 0,575    | 6,1349e-004   | 17,653        | 0,41719       |
| 0,675    | 3,7179e-003   | 18,355        | 0,53351       |
| 0,775    | 6,5813e-004   | 19,263        | 0,47942       |
| 0,875    | 1,1718e-003   | 19,128        | 0,50171       |
| 0,9375   | 1,2517e-003   | 19,617        | 0,52787       |
| 1,       | 1,042e-003    | 20,897        | 0,57006       |

**TABLE 20****Model (B4) > Static Structural (B5) > Solution (B6) > Stress Safety Tools**

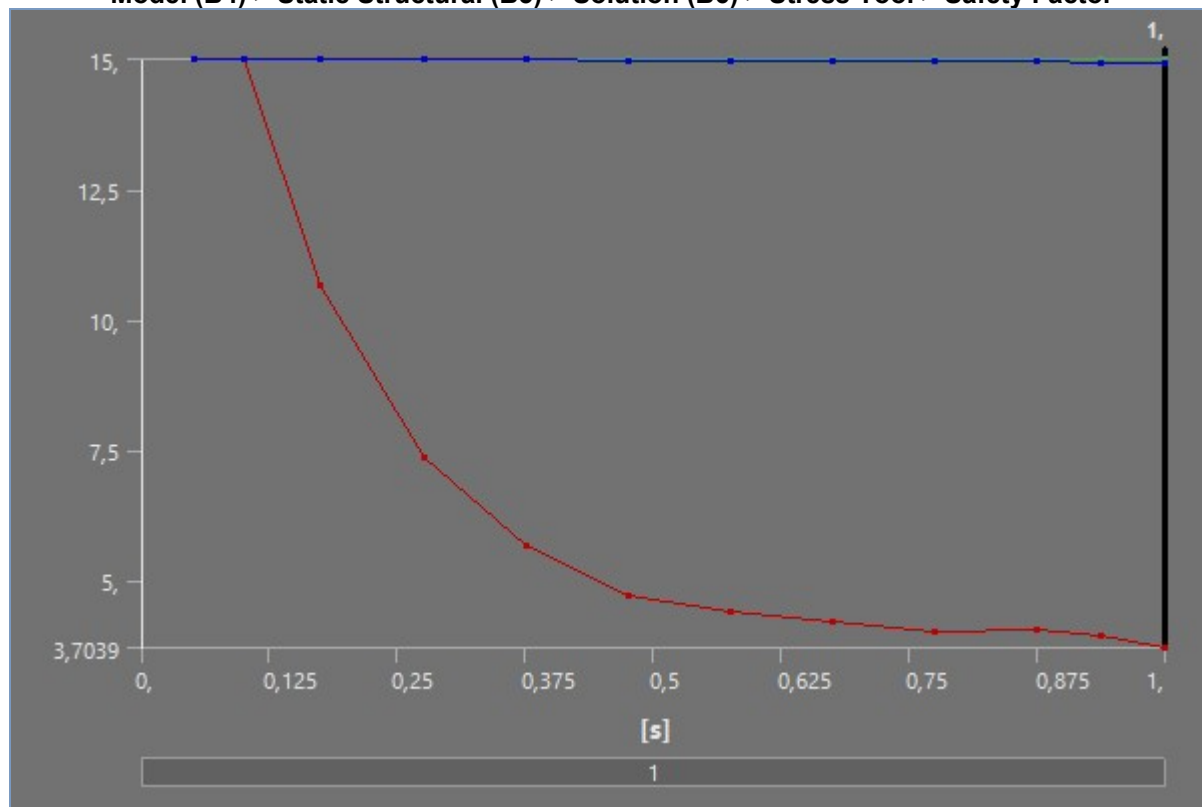
|                   |                            |
|-------------------|----------------------------|
| Object Name       | <i>Stress Tool</i>         |
| State             | Solved                     |
| <b>Definition</b> |                            |
| Theory            | Max Equivalent Stress      |
| Stress Limit Type | Tensile Yield Per Material |

**TABLE 21****Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Results**

|                   |                      |
|-------------------|----------------------|
| Object Name       | <i>Safety Factor</i> |
| State             | Solved               |
| <b>Scope</b>      |                      |
| Scoping Method    | Geometry Selection   |
| Geometry          | All Bodies           |
| <b>Definition</b> |                      |
|                   |                      |

| Type                             | Safety Factor                |
|----------------------------------|------------------------------|
| By                               | Time                         |
| Display Time                     | Last                         |
| Separate Data by Entity          | No                           |
| Calculate Time History           | Yes                          |
| Identifier                       |                              |
| Suppressed                       | No                           |
| <b>Integration Point Results</b> |                              |
| Display Option                   | Averaged                     |
| Average Across Bodies            | No                           |
| <b>Results</b>                   |                              |
| Minimum                          | 3,7039                       |
| Minimum Occurs On                | max_pressure Вырез-Вытянуть2 |
| <b>Minimum Value Over Time</b>   |                              |
| Minimum                          | 3,7039                       |
| Maximum                          | 15,                          |
| <b>Maximum Value Over Time</b>   |                              |
| Minimum                          | 15,                          |
| Maximum                          | 15,                          |
| <b>Information</b>               |                              |
| Time                             | 1, s                         |
| Load Step                        | 1                            |
| Substep                          | 12                           |
| Iteration Number                 | 39                           |

**FIGURE 7**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor**



**TABLE 22**  
**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor**

| Time [s] | Minimum | Maximum | Average |
|----------|---------|---------|---------|
| 5,e-002  | 15,     |         | 15,     |
| 0,1      |         |         |         |

|        |        |     |        |
|--------|--------|-----|--------|
| 0,175  | 10,645 | 15, | 14,999 |
| 0,275  | 7,3569 |     | 14,991 |
| 0,375  | 5,6498 |     | 14,983 |
| 0,475  | 4,6934 |     | 14,966 |
| 0,575  | 4,3845 |     | 14,958 |
| 0,675  | 4,2168 |     | 14,944 |
| 0,775  | 4,018  |     | 14,943 |
| 0,875  | 4,0464 |     | 14,944 |
| 0,9375 | 3,9456 |     | 14,937 |
| 1,     | 3,7039 |     | 14,921 |

## Material Data

### PA12-CF10

**TABLE 23**  
**PA12-CF10 > Constants**

|                                  |                               |
|----------------------------------|-------------------------------|
| Density                          | 1,04e-006 kg mm <sup>-3</sup> |
| Coefficient of Thermal Expansion | 7,e-005 K <sup>-1</sup>       |

**TABLE 24**  
**PA12-CF10 > Color**

|      |       |      |
|------|-------|------|
| Red  | Green | Blue |
| 235, | 222,  | 222, |

**TABLE 25**  
**PA12-CF10 > Isotropic Elasticity**

| Young's Modulus MPa | Poisson's Ratio | Bulk Modulus MPa | Shear Modulus MPa | Temperature K |
|---------------------|-----------------|------------------|-------------------|---------------|
| 3311,               | 0,38            | 4598,6           | 1199,6            | 295,15        |
| 1500,               | 0,4             | 2500,            | 535,71            | 343,15        |

**TABLE 26**  
**PA12-CF10 > Tensile Yield Strength**

|                            |
|----------------------------|
| Tensile Yield Strength MPa |
| 77,4                       |

**TABLE 27**  
**PA12-CF10 > Compressive Ultimate Strength**

|                                   |
|-----------------------------------|
| Compressive Ultimate Strength MPa |
| 90,                               |